Team M

grade: 87
comments: no use-case diagram was given, which is crucial to organize the actors and use cases with clean overall view. The use cases are described fine, but more details can be given instead of over-concise descriptions. A fine report overall.

College0 Software Requirements Specification For Subsystem

Version 1.0

Revision History

Date	Version	Description	Author
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Software Requirements Specification

1. Introduction

This document explains what the College0 system is supposed to do. It lists all the features and rules for building the software. The goal is to help the development team understand exactly what to build and help the testers know what to check

1.1 Purpose

The purpose of this paper is to write down all the rules and features for College0. We want to make sure there is no confusion about what the system should do. This document will be used by everyone working on the project to stay on the same page.

1.2 Scope

This document covers College0, which is a software application for managing a small online college program. The system handles the following:

- User management for four types of users: Registrars, Instructors, Students, and Visitors
- Semester management with four periods: Class Set-up, Course Registration, Class Running, and Grading Period
- Course registration and waitlist management
- Grade assignment and academic standing tracking
- Review and rating system for classes
- Complaint handling and warning system
- An AI-powered assistant that answers user questions

This is a small system meant to demonstrate how the features work. It will run as a local application on a computer.

1.3 Definitions, Acronyms, and Abbreviations

Term	Meaning
Registrar	The administrator user who has full control over the system and can perform any action.
Instructor	A teacher who is assigned to one or more classes. Instructors can manage waitlists, assign grades, and view student records.
Student	A user who is enrolled in the college. Students can register for classes, write reviews, view grades, and apply for graduation.
Visitor	An unauthenticated user who can only view public information and apply to become a student or instructor.

Term	Meaning
GPA	**Grade Point Average.** A numerical value that represents a student's academic performance on a scale of 0 to 4.
LLM	**Large Language Model.** A type of artificial intelligence that can understand and respond to human language. Examples include ChatGPT.
GUI	**Graphical User Interface.** The visual screen where users interact with the system using buttons, text boxes, and menus.
SRS	**Software Requirements Specification.** This document.

1.4 References

N/A

1.5 Overview

The remainder of this document is organized as follows:

- **Section 2** provides an overall description of the system, including the types of users, main tasks, and key assumptions.
- **Section 3** contains the detailed requirements, organized into use-case reports and supplementary requirements.
- **Section 4** describes the supporting information, including appendices that will be created later.

2. Overall Description

This section provides a general overview of the College0 system. It describes who will use the system and what factors affect how the system works. Specific details are saved for Section 3.

2.1 Use-Case Model Survey

The College0 system has four types of users, called actors. Each actor can perform different tasks within the system.

Actors Table:

Actor	Description
Visitor	An unauthenticated user. They can view public information (top/bottom rated classes, top GPA students) and submit applications to become a Student or Instructor.
Student	An enrolled user with an active account. They can register for classes, write reviews, track academic records, apply for graduation, and file complaints.
Instructor	A faculty member assigned to classes. They manage rosters and waitlists, assign final grades, view student records, and file complaints.
Registrar	The system administrator. They manage applications, semester stages, and class setups, process complaints, and oversee warnings or suspensions.

Main Use Cases:

Use Case	Description
View Public Info	All users can view the dashboard featuring college intros, class ratings, and top student rankings.
Apply for Membership	Visitors submit applications; Registrars review and approve/deny them for Student or Instructor roles.
Manage Account	Users can log in and change passwords. Mandatory password change is required for new students.
Manage Semester	The Registrar transitions the system through: *Setup, Registration, Running,* and *Grading* periods.
Class Registration	Students enroll in 2–4 classes, subject to capacity and schedule conflict checks.

Use Case	Description
Manage Waitlist	Instructors manually move students from the waitlist into active rosters as seats open up.
Cancel Small Classes	Automated system task: Cancels classes with <3 students and triggers a special registration window.
Write Reviews	Students provide 1–5 star ratings and comments. Includes automated language filtering and anonymity.
Assign Grades	Instructors input final grades; system flags missing grades or unusual distributions.
Track Standing	Automated GPA calculation, Honor Roll placement, and issuance of academic warnings/terminations.
Apply for Graduation	Available to students with 8+ completed classes. Registrar performs final verification and approval.
Handle Complaints	Students/Instructors file reports; Registrar investigates and issues warnings to responsible parties.
Warnings & Fines	The system tracks user warnings. A 3-warning limit triggers a one-semester suspension and a fine.
Use AI Assistant	Users query a text box. System checks local knowledge first, then reverts to external AI if needed.
Personal Dashboard	Role-based landing page showing customized data and tasks upon successful login.

2.2 Assumptions and Dependencies

The following assumptions and dependencies affect the development and operation of the College0 system.

Item	Description
Platform	The system is a **local desktop application**. It is not required to be web-based or accessible via the internet for core operations.
User Scale	The system is designed for a small-scale demonstration, tested with a dataset of approximately **10 students**.
Connectivity	An **active internet connection** is mandatory specifically for the AI assistant feature to reach external services.
External AI	The AI assistant relies on an **External LLM API**. The system must be configured to send and receive data from this third-party service.
Content Moderation	A **"Bad Words List"** must be manually configured by the Registrar during setup; the system does not include a pre-loaded database of taboo words.

3. Specific Requirements

This section contains the detailed requirements for the College0 system. Each requirement is clearly labeled to help developers and testers.

3.1 Use-Case Reports

The following use-case reports describe the detailed steps and rules for each main task in the system.

UC-1: Apply to Become a Student

Element	Description
Actor	Visitor, Registrar
Precondition	Visitor is not currently a student.
Main Flow	1. Visitor submits application. 2. System auto-accepts if GPA > 3.0 and quota is available. 3. Registrar reviews remaining. 4. Upon approval, system generates ID and temporary password.
Postcondition	Visitor becomes a student or receives a rejection notice with a reason.

UC-2: Apply to Become an Instructor

Element	Description
Actor	Visitor, Registrar
Precondition	Visitor is not currently an instructor.
Main Flow	1. Visitor submits instructor application. 2. Registrar reviews and approves/rejects. 3. If approved, the user is added to the instructor pool for class assignment.
Postcondition	Visitor becomes an instructor or is rejected (no reason required).

UC-3: Set Up Classes

Element	Description
Actor	Registrar
Precondition	System state is in **Class Set-up** period.
Main Flow	1. Registrar adds new class. 2. Enters name, schedule, instructor, and max capacity. 3. System saves data.
Postcondition	Classes are created and editable before registration begins.

UC-4: Register for Classes

Element	Description
Actor	Student
Precondition	**Registration Period** is active; Student is not suspended.
Main Flow	1. Student selects 2–4 classes. 2. System validates capacity and time conflicts. 3. Student is enrolled or added to waitlist.

Element	Description
Alternative	**Retaking:** Allowed only if previous grade was "F".

UC-5: Manage Waitlist

Element	Description
Actor	Instructor
Precondition	**Class Running** period; Class has a waitlist.
Main Flow	1. Instructor views waitlist. 2. Instructor selects a student. 3. System verifies seat availability and moves student to the active roster.
Postcondition	Student is enrolled and roster is updated.

UC-6: Monitor Class Health (System Automatic)

Element	Description
Actor	System (Automatic)
Precondition	**Class Running** period.
Main Flow	1. System scans enrollment. 2. Classes with < 3 students are canceled. 3. Special registration opens for affected students. 4. Instructor receives a warning.
Postcondition	Small classes removed; potential instructor suspension if all their classes are cut.

UC-7: Write a Review

Element	Description
Actor	Student
Precondition	Student is enrolled; final grades are not yet posted.
Logic	**0 Taboo:** Published. **1-2 Taboo:** Censored with * + 1 warning. **3+ Taboo:** Blocked + 2 warnings.
Postcondition	Review is recorded; identity is hidden from all except the Registrar.

UC-8: Assign Grades

Element	Description
Actor	Instructor
Precondition	**Grading Period** is active.
Main Flow	1. Instructor submits grades. 2. System validates completion. 3. Instructor flagged if GPA is outlier (>3.5 or <2.5).
Postcondition	Grades recorded; warnings issued for missing submissions.

UC-9: Track Student Standing

Element	Description
Actor	System (Automatic)
Precondition	Grades have been finalized for the semester.
Main Flow	1. GPA calculated. 2. < 2.0 = Termination. 3. 2.0–2.25 = Warning + Interview. 4. > 3.75 = Honor Roll.

Element	Description
Postcondition	Academic status updated; Honor Roll removes one warning.

UC-10: Apply for Graduation

Element	Description
Actor	Student, Registrar
Precondition	Student has completed 8 classes.
Main Flow	1. Student applies. 2. Registrar verifies credits. 3. If met, account deactivated and student graduates.
Alternative	**Reckless Application:** Warning issued if requirements are not met.

UC-11: File a Complaint

Element	Description
Actor	Student, Instructor, Registrar
Precondition	User is authenticated.
Main Flow	1. User submits form. 2. Registrar investigates. 3. Registrar issues warning or takes no action.
Postcondition	Resolution recorded; Registrar may warn instructor if complaint was frivolous.

UC-12: Track Warnings and Suspensions

Element	Description
Actor	System (Automatic), Registrar
Precondition	A user receives a warning.
Main Flow	1. Warning count increments. 2. At 3 warnings, system triggers automatic one-semester suspension.
Postcondition	User must pay a fine to restore login access after suspension.

UC-13: View Personal Dashboard

Element	Description
Actor	All Roles
Precondition	User is logged in.
Main Flow	1. System identifies role. 2. Displays relevant data (Grades for Students, Rosters for Instructors, All for Registrar).
Postcondition	Role-specific tutorial displayed for new students.

UC-14: Use AI Assistant

Element	Description
Actor	Visitor, Student, Instructor
Precondition	Internet connection available.
Main Flow	1. User asks question. 2. Local vector DB searched first. 3. External LLM used as fallback with inaccuracy warning.

Element	Description
Postcondition	Response provided based on user's role-based permissions.

3.2 Supplementary Requirements

The following requirements are not covered in the use cases but are necessary for the complete system.

ID	Category	Requirement Description
UI-1	**User Interface**	The system shall use a single main window. No pop-up windows shall appear on top of each other.
UI-2	**User Interface**	The main screen shall display: a welcome message, college introduction, top 3 highest rated classes, top 3 lowest rated classes, and top 3 students by GPA.
UI-3	**User Interface**	Buttons and text shall be sized for high readability and ease of interaction (clickable).
SEC-1	**Security**	New students shall be issued a unique student ID and a temporary initial password.
SEC-2	**Security**	Users must be forced to change their initial password upon their first successful login.
SEC-3	**Security**	Passwords shall be stored using secure hashing to prevent unauthorized access.
SEC-4	**Security**	Role-Based Access Control (RBAC) shall ensure users only access functions appropriate for their role (e.g., students cannot view others' grades).
AI-1	**AI Assistant**	The system shall maintain a local vector database containing college-specific information.
AI-2	**AI Assistant**	The system shall prioritize searching the local database before querying an external LLM.

ID	Category	Requirement Description
AI-3	**AI Assistant**	Information retrieved from the local database shall be displayed as verified data (no warning).
AI-4	**AI Assistant**	Responses generated by external LLMs must include a clear disclaimer regarding potential inaccuracies.
CF-1	**Creative Feature**	A Schedule Optimizer tool shall be available to the Registrar during the Class Set-up period.
CF-2	**Creative Feature**	The tool shall analyze historical data to suggest times that minimize scheduling conflicts.
CF-3	**Creative Feature**	The system shall flag warnings if two frequently co-registered classes are scheduled simultaneously.
CF-4	**Creative Feature**	The tool shall provide recommended class capacities based on historical enrollment trends.
CF-5	**Creative Feature**	The Registrar shall have final authority to accept, modify, or ignore optimizer suggestions.

4. Supporting Information

The following supporting materials will be created to accompany this document. These materials help explain the system in more detail.

Item	Description	Part of Requirements?
Appendix A: Use-Case Diagram	A visual diagram mapping all **four actors** to their specific system tasks.	**Yes**
Appendix B: Screen Mockups	Wireframes/Mockups showing the single-window layout for each user role (GUI design).	**Yes**

Item	Description	Part of Requirements?
Appendix C: Data Dictionary	A detailed list of data objects (Student, Course, Grade, etc.) and their specific attributes.	**Yes**
Appendix D: Sample Dataset	A set of 10 sample students and 5 classes used for testing and the final demonstration.	**No** (Testing only)